

2 Systems

2.1 Basic Concepts in Vibrations

The study of vibrations, within the broader field of classical mechanics, is the investigation of oscillations that occur about an equilibrium point. Vibrations, both desired and undesired, are present in all mechanical systems and can be helpful (e.g. a soil sieve, rotary sander) or destructive (e.g. an aircraft frame in resonance). The oscillations that form a vibrating system may be periodic (e.g., pendulum) or random (e.g. turbulence in an airplane), or a combination of the two.

Vibrations impact our daily lives in a variety of ways, from the sound made by banjo strings that vibrates between 140 and 400 Hz to the 4-6 Hz vibration felt by a passenger in a car seat. The consideration of the vibrations and their associated mathematical modeling are an important factor in the design of mechanical systems. In the material that follows, the fundamental theories of vibration are presented and modeled using fundamental physical principles such as Newton's three laws of motion. These models are analyzed using the mathematical tools of calculus and differential equations.

Review 2.1 Newton's three laws of motion:

1. In an inertial frame of reference, an object either remains at rest or continues to move at a constant velocity, unless acted upon by a force.
2. In an inertial reference frame, the vector sum of the forces F on an object is equal to the mass m of that object multiplied by the acceleration of the object: $F = ma$. (It is assumed here that the mass m is constant)
3. When one body exerts a force on a second body, the second body simultaneously exerts a force equal in magnitude and opposite in direction on the first body.

2.2 Single Degree-of-Freedom Systems

In its simplest form, the phenomenon of vibration is the exchange of energy between potential and kinetic energy. Therefore, a vibrating system must have a component that stores potential energy. This component must also be capable of releasing the energy as kinetic energy. This kinetic energy is stored in the movement of a mass where the measure of this movement is the velocity of the system and the continuous interchange between potential and kinetic energy is the vibration of the system. The simplest vibrating systems can be modeled as a single-degree-of-freedom (1-DOF) system. In a 1-DOF system, one variable can describe the motion of a system. Potential examples of 1-DOF systems include:

1. yo-yo
2. pogo stick
3. door swinging on axis

4. throttle (gas pedal)

Variables often used for describing 1-DOF systems are $x(t)$, $y(t)$, $z(t)$, and $\theta(t)$. Examples of 1-DOF systems are presented in figure 2.1 where the assumption of small displacements is made. Note: we will often drop the “(t)” for simplicity in this material.

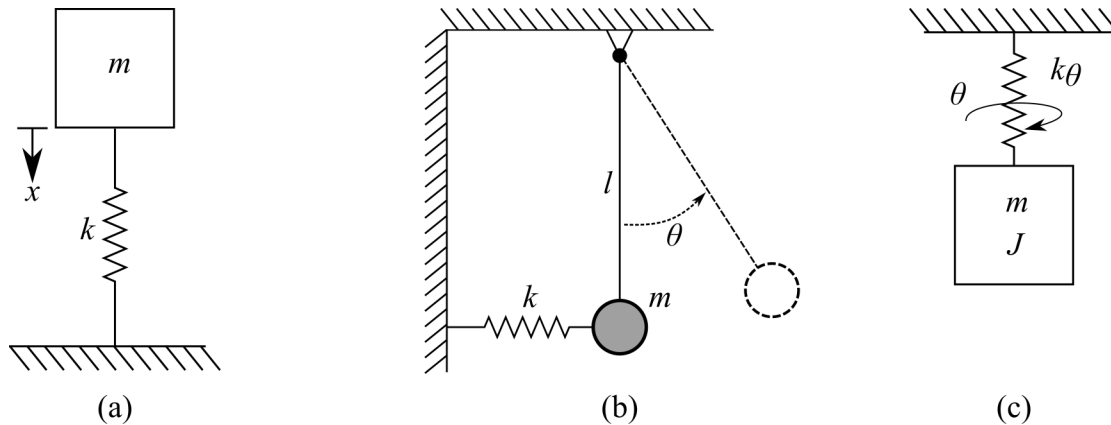


Figure 2.1: Examples of single degree of freedom (DOF) systems showing: (a) a vertical spring-mass system; (b) a simple pendulum; and (c) a rotational spring-mass system.

2.2.1 Spring-Mass Model

“All models are wrong, but some are useful”

George E.P. Box (1919 - 2013)

Newtonian physics describes the motion of particles in terms of displacement x , velocity $\dot{x}$, and acceleration $\ddot{x}$ vectors. Moreover, from Newton’s second law of motion says that the change in the velocity of mass in motion is a product of the force acting on the mass. A simple way to express this phenomenon is through a spring-mass model as presented in figure 2.2. These spring-mass models neglect the mass of the spring and concentrate all the mass of the system into a single point. Note that in this case the force vector and mass-acceleration vectors lie on the same axis and as such are collinear. Therefore, these vectors can be easily treated as scalars simplifying the math used in the modeling of the system.

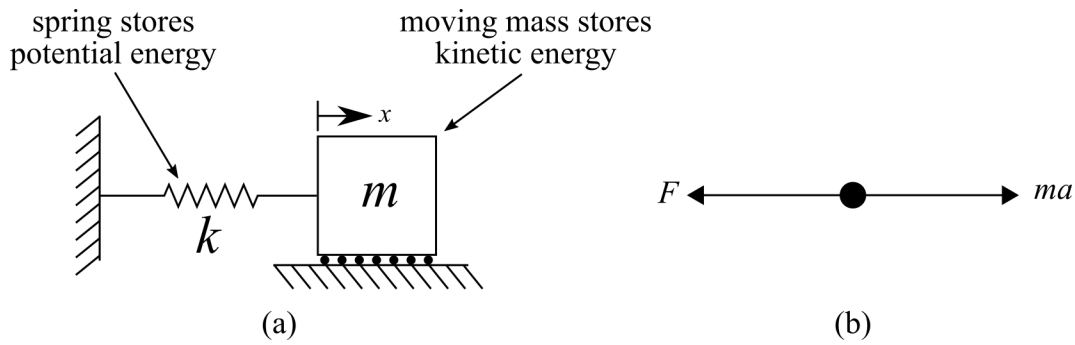


Figure 2.2: A single-degree-of-freedom (1-DOF) spring mass model showing: (a) annotated schematic of a mass-spring system; and (b) the equivalent free-body diagram represented as a point-mass system.

2.2.2 Linear Springs

Springs are mechanical devices that store energy, moreover, ideal spring is a theoretical representation of this mechanical device that is massless and responds with a linear increase in force for a unit increase in displacement (i.e. $F = kx$). For simplicity, the spring in the spring-mass model considered here is assumed always ideal linear springs. A graphical representation of the idealized linear spring is presented in figure 2.3 where a unit force F applied to the free end of the spring results in a unit displacement x of the spring. The resulting mathematical relationships, $F = kx$, is known as Hooke's Law. Nonlinear springs add considerable complexity to the modeling of spring-mass systems, therefore, these are not considered in this introductory work.

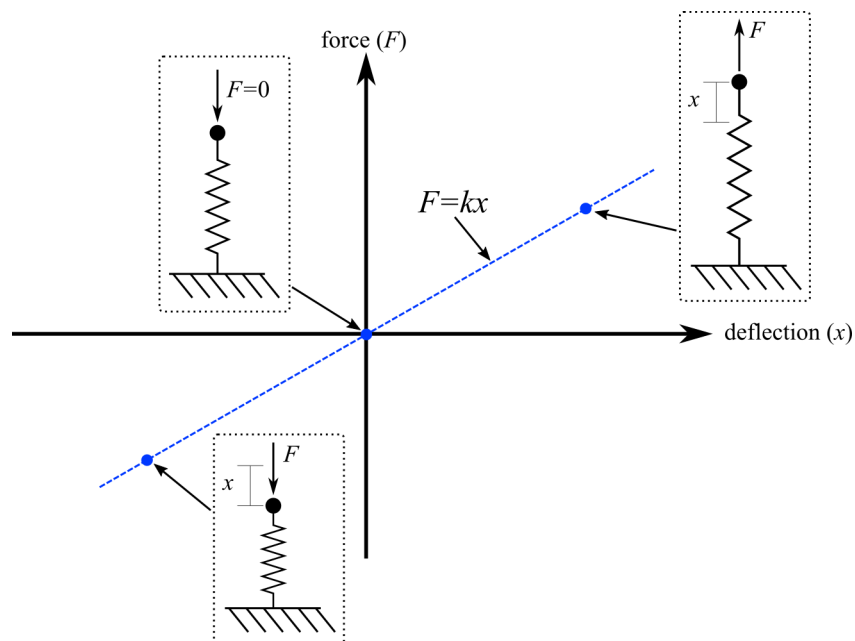


Figure 2.3: Force-displacement plot for a linear spring.

2.2.3 Equation of Motion for an Oscillating System

An Equation of Motion (EOM) is an equation that provides a basis for modeling a vibrating system about its equilibrium point and relates the transfer of the potential energy from the spring to the kinetic energy mass. In developing the EOM we assume that any surfaces are frictionless and as such, no energy is extracted from the vibrating system. Referencing the 1-DOF system in figure 2.4(a), and assuming the mass only moves in the x direction, the only force acting on the mass in the x direction is the force that results from the elongation of the spring as annotated in figure 2.4(b). Therefore, the sum of forces in along the x axis must equal the mass (m) times the acceleration of the mass ($a\ddot{x}$).

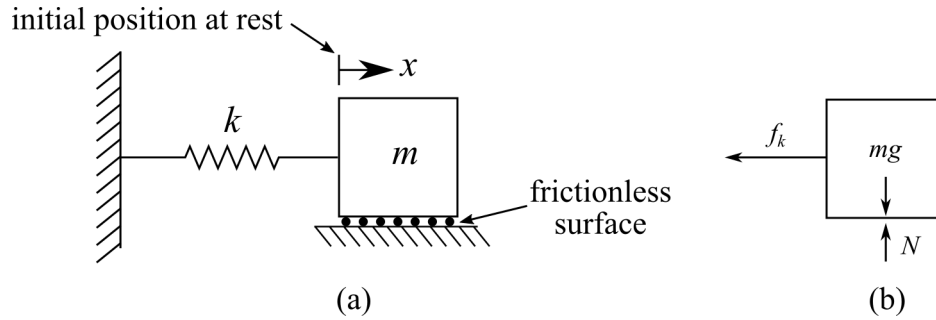


Figure 2.4: A spring mass model of a 1-DOF system showing: (a) a schematic of the system; (b) free-body diagram of the system at its initial position.

Considering that positive displacements are to the right, the standard form of the equation of motion for an undamped system without any excitation is expressed as:

$$s_1\ddot{x} + s_2x = 0 \quad (1)$$

where s_1 and s_2 are constants to be determined for the specific system. A systematic approach to obtaining free-body diagram (FBD) of a system under vibration can be expressed in three steps:

1. Draw a free-body diagram (FBD) at the system's equilibrium and displaced position (without a displacing force).
2. Apply Newton's second law to both FBDs (equilibrium and displaced).
3. Combine the equations to write the EOM in standard form with the forcing component on the right-hand side. For free vibration, the forcing component is 0.

Solving these three steps for 1-DOF system presented in figure 2.4 results in the EOM:

$$m\ddot{x} + kx = 0 \quad (2)$$

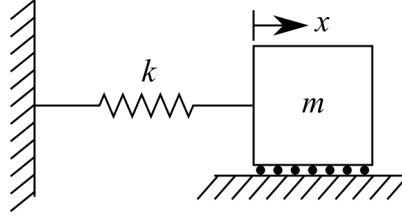
Review 2.2 A second-order linear homogeneous differential equation has the form:

$$a\ddot{x} + b\dot{x} + cx = 0 \quad (3)$$

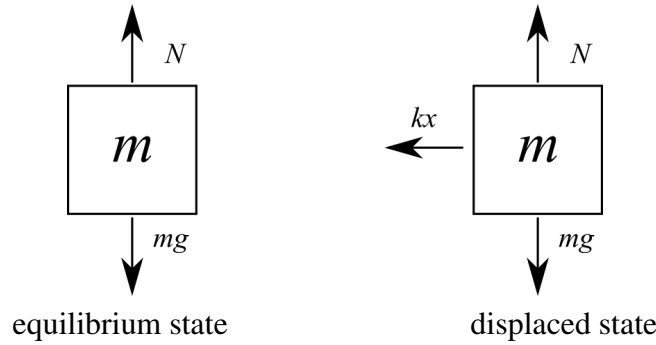
The EOM for a 1-DOF system under a free vibration is a second-order differential equation due to acceleration ($\ddot{x}$) being the second derivative of displacement (x) and homogeneous as

the forcing function (right-hand side of the equations) is zero. In EOM's current form, $m = k$, $b = 0$, and $c = k$. In future work, b will account for damping in the vibrating system.

Example 2.1 Considering the system:



Step-1 Define the direction of displacement, and draw the FBD for the equilibrium and displaced state.



The equation for the equilibrium state is:

$$\sum \rightarrow F_x = 0 \quad (4)$$

and in the displaced state:

$$\sum \rightarrow F_x = -kx \quad (5)$$

This equation does not equal zero as the FBD does not account for the restoring force.

Step-2 Apply Newton's second law (we want to store energy in the kinetic state) of motion to the sum of forces for the displaced position we get:

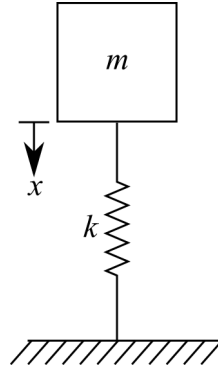
$$ma = m\ddot{x} = \sum \rightarrow F_x = -kx \quad (6)$$

$$m\ddot{x} = -kx \quad (7)$$

Step-3 Rearrange in the Equation to construct an EOM:

$$m\ddot{x} + kx = 0 \quad (8)$$

Example 2.2 Some systems will have an initial displacement, as the system will oscillate around this position we need to define the EOM about this position. Considering the system:



Step-1 Define the direction of displacement, and draw the FBD for the equilibrium and displaced state.



The equation for the equilibrium state is:

$$+\downarrow \sum F_x = mg - k\delta = 0 \quad (9)$$

and in the displaced state:

$$+\downarrow \sum F_x = mg - k(\delta + x) \quad (10)$$

This equation does not equal zero as the FBD does not account for the restoring force.

Step-2 Apply Newton's second law (we want to store energy in the kinetic state) of motion to the sum of forces for the displaced position we get:

$$m\ddot{x} = +\downarrow \sum F_x = mg - k\delta - kx \quad (11)$$

We can then use the information from the equilibrium state to cancel out some terms, this becomes:

$$m\ddot{x} = -kx \quad (12)$$

Step-3 Rearrange in the Equation to construct an EOM:

$$m\ddot{x} + kx = 0 \quad (13)$$

2.3 Forcing Function

2.3.1 Step Function

A step function is a common loading situation that can represent the sudden application of a load, such as dropping a load into a truck, a car going over a curb, or a motor starting up. The step function, also known as the *Heaviside function*, is denoted by $\Phi(t)$ and is defined as

$$\Phi(t) = \begin{cases} 1, & t > 0, \\ 0, & t < 0. \end{cases} \quad (14)$$

The step function represents an input that changes instantaneously from zero to a constant value at $t = 0$. The value assigned exactly at $t = 0$ does not affect the applications considered here.

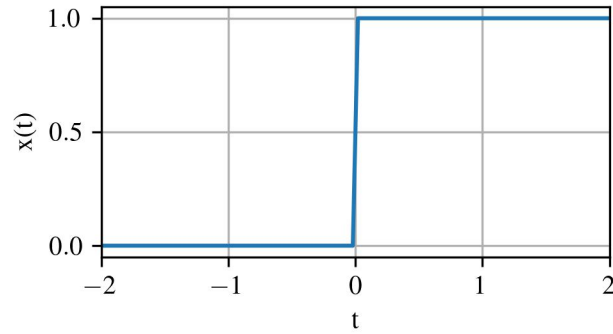


Figure 2.5: Step function.

2.3.2 Pulse Function

The pulse function $\rho(t; \tau)$ consists of a step up at $t = 0$ followed by a step down at $t = \tau$. The parameter τ defines the duration of the pulse. The amplitude is chosen as $1/\tau$ so that the area under the pulse is constant and equal to unity ($A = 1$):

$$A = \int_0^\tau \frac{1}{\tau} dt = 1. \quad (15)$$

The pulse function is therefore defined as

$$\rho(t; \tau) = \begin{cases} 1/\tau, & 0 < t < \tau, \\ 0, & \text{else.} \end{cases} \quad (16)$$

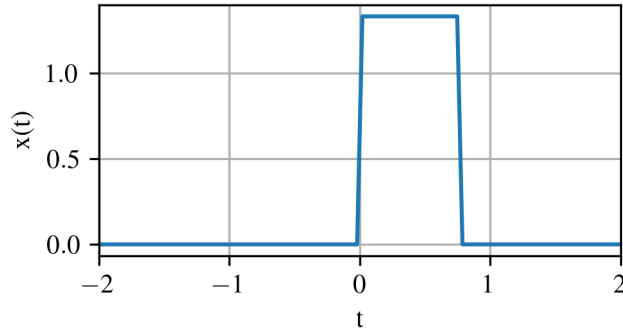


Figure 2.6: Pulse function.

2.3.3 Impulse Function

Shock loads on mechanical systems represent a very common source of vibration. These short-duration forces are also called impulses. An impulse excitation is defined as a force that is applied for a very short, or infinitesimal, length of time. An impulse is a nonperiodic force that is represented by the symbol δ . The impulse function $\delta(t)$ is also known as the “Dirac function”, “Dirac delta function”, “Dirac impulse function”, or “delta function”.

The impulse function is obtained from the pulse function $\rho(t; \tau)$ by letting the pulse duration τ become infinitesimally small:

$$\delta(t) = \lim_{\tau \rightarrow 0} \rho(t; \tau). \quad (17)$$

As the pulse duration approaches zero, its amplitude increases such that the area under the function remains equal to unity. Thus, the impulse satisfies

$$A = \int_{-\infty}^{\infty} \delta(t) dt = 1. \quad (18)$$

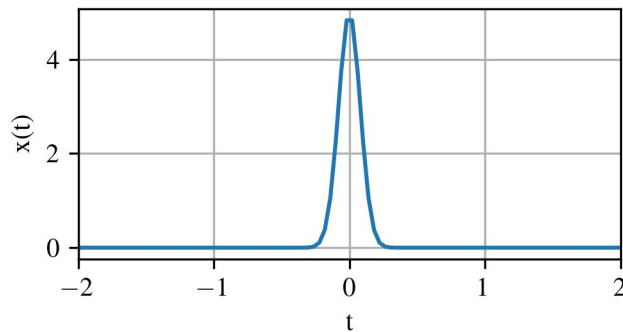


Figure 2.7: Impulse function.

2.3.4 Ramp Function

The ramp function represents an input that begins at zero and increases linearly with time. As shown in Figure 2.8, the function is zero before $t = 0$ and increases with a constant slope after

$t = 0$. The ramp function is defined as

$$f(t) = \begin{cases} t, & t > 0, \\ 0, & t < 0. \end{cases} \quad (19)$$

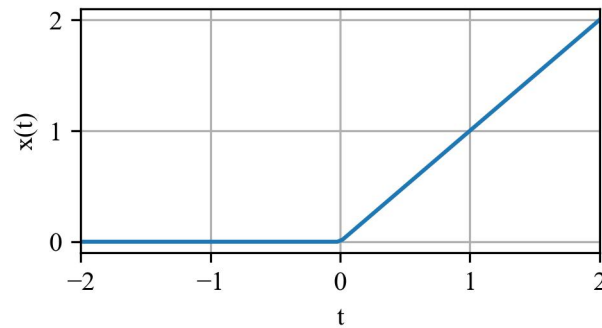


Figure 2.8: Ramp function.

2.4 Introduction to Stability

The stability of a system is explained through “poles” and “zeros”. The poles and the zeros of a system determine whether the system is stable, and how well the system performs.

1. A system at a pole has an output that is infinite even though the input to the system was finite (i.e. unstable).
2. A system at a zero has an output that is finite even though the input to the system was infinite (i.e. stable).

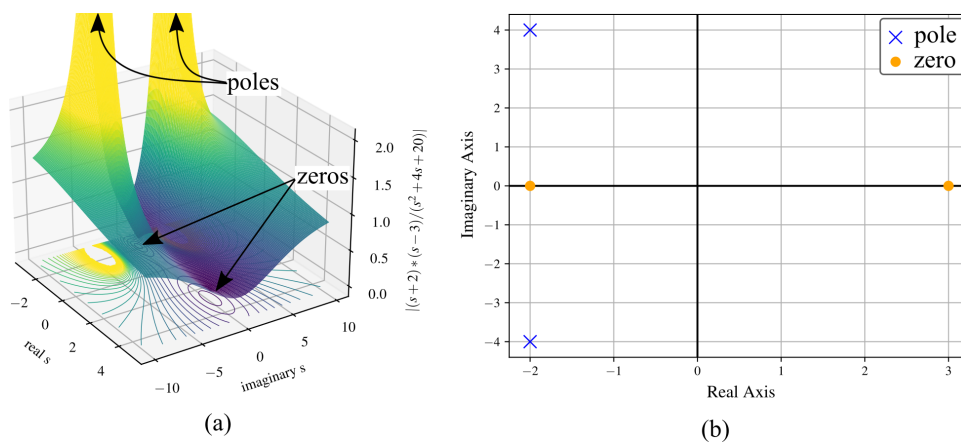


Figure 2.9: Poles and zeros.

Control systems can be designed simply by assigning specific values to the poles and zeros of the system. Physically realizable control systems must have a number of poles greater than the

number of zeros. Systems that satisfy this relationship are called Proper. For a rational polynomial transfer function: The poles of a transfer function are defined as the roots of the denominator polynomial of the transfer function. The zeros of a transfer function are simply the roots of the numerator polynomial. Visualizing of poles and zeros in the complex space can be helpful in understanding the stability of the system. Figure 2.9 shows the complex space for an arbitrary transfer function $(s+2)(s-3)/s^2+4s+20$ where figure 2.9(a) shows the 3D space while figure 2.9(b) just reports the poles and zeros of the transfer function. Figure 2.10 reports the transfer functions for 3 additional transfer functions.

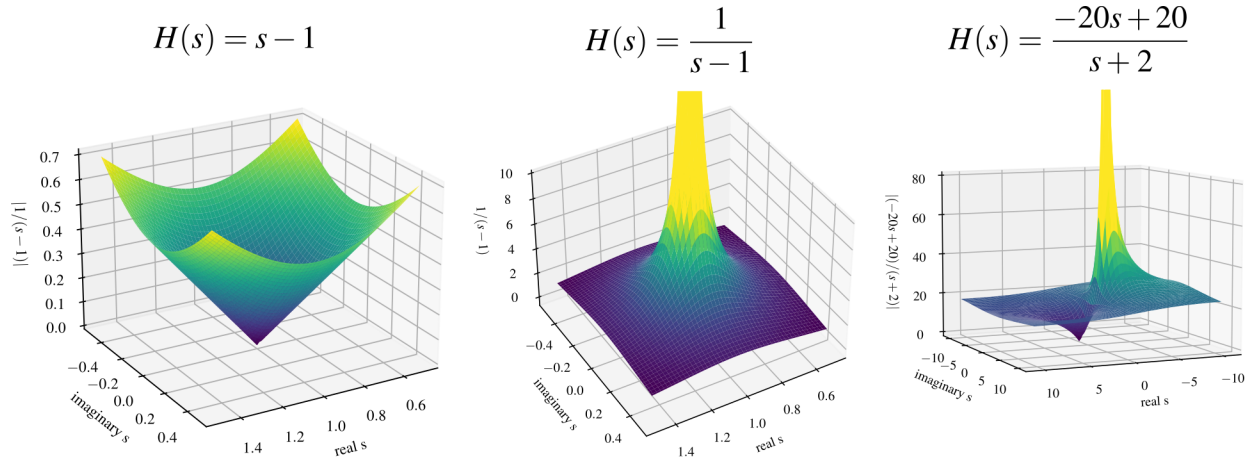


Figure 2.10: Poles and zeros in the complex space, plotted for 3 transfer functions.

2.5 1st Order Systems

2.5.1 Spring-damper mechanism

Consider the system:

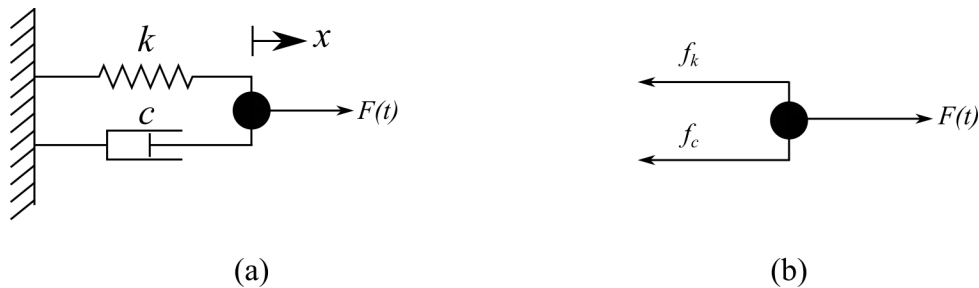


Figure 2.11: Damped 1-DOF system with an external force ($F(t)$) applied, showing: (a) the system configuration; and (b) the free body diagram

Making the assumption that this system does not have any mass, the force balance equation is written as:

$$-kx - c\dot{x} + f^* = 0 \quad (20)$$

this becomes the equation of motion

$$c\dot{x} + kx = f^* \quad (21)$$

normalizing the equation of motion by the stiffness k ,

$$\frac{c}{k}\dot{x} + x = \frac{1}{k}f^* \quad (22)$$

define the *time constant* T as

$$T = \frac{c}{k} \quad (23)$$

and the normalized forcing function $f(t) = \frac{1}{k}f^*(t)$. This results in the 1st-order ODE in standard form:

$$T\dot{x}(t) + x(t) = f(t) \quad (24)$$

A general solution for this ODE is written as:

$$x(t) = x_c(t) + x_p(t) \quad (25)$$

where: x_c is called the “complementary solution” and satisfies the homogeneous equation and satisfies the homogeneous equation

$$T\dot{x}_c + x_c = 0 \quad (26)$$

The homogeneous equation is obtained from equation 24 by making zero the right hand side. The complementary solution x_c is the free response.

$x_p(t)$ is called the “particular solution” and is used to satisfy the right-hand side of equation 24. Substituting equation 25 into equation 24 yields:

$$T(\dot{x}_c + \dot{x}_p) + x_c + x_p = f(t) \quad (27)$$

or

$$T\dot{x}_c + x_c + T\dot{x}_p + x_p = f(t) \quad (28)$$

As $T\dot{x}_c + x_c = 0$:

$$T\dot{x}_p + x_p = f(t) \quad (29)$$

This shows that:

- $T\dot{x}_c + x_c$ is the solution to the free response.
- $T\dot{x}_p + x_p$ is the solution to the forced response.

2.5.2 Homogeneous Equation (Free Response)

The Homogeneous equation is obtained from equation 24 and setting the right hand side to zero:

$$T\dot{x}_c + x_c = 0 \quad (30)$$

To solve, assume:

$$\begin{aligned} x_c(t) &= ce^{pt} \\ \dot{x}_c(t) &= pce^{pt} \end{aligned} \quad (31)$$

Inserting equation 31 into equation 30 gives:

$$T(pce^{pt}) + ce^{pt} = 0 \quad (32)$$

canceling out like terms results in the “characteristic equation”:

$$Tp + 1 = 0 \quad (33)$$

therefore:

$$p = \frac{-1}{T} \quad (34)$$

where this is a “pole” of the system. Inserting equation 34 into equation 31 results in the “free response” of the system:

$$x_c(t) = ce^{-t/T} \quad (35)$$

as shown in figure 2.12. Note that the lack of mass in the system prevents it from oscillate as there is no way for it to hold kinetic energy, and therefore no way for it to exchange energy between potential and kinetic forms.

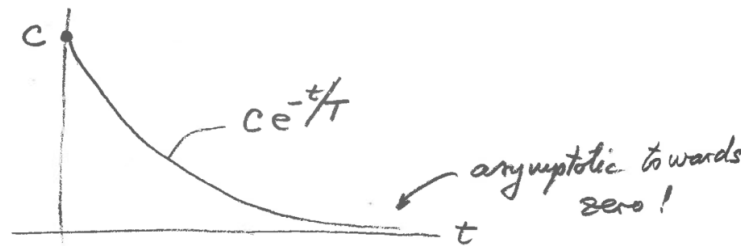


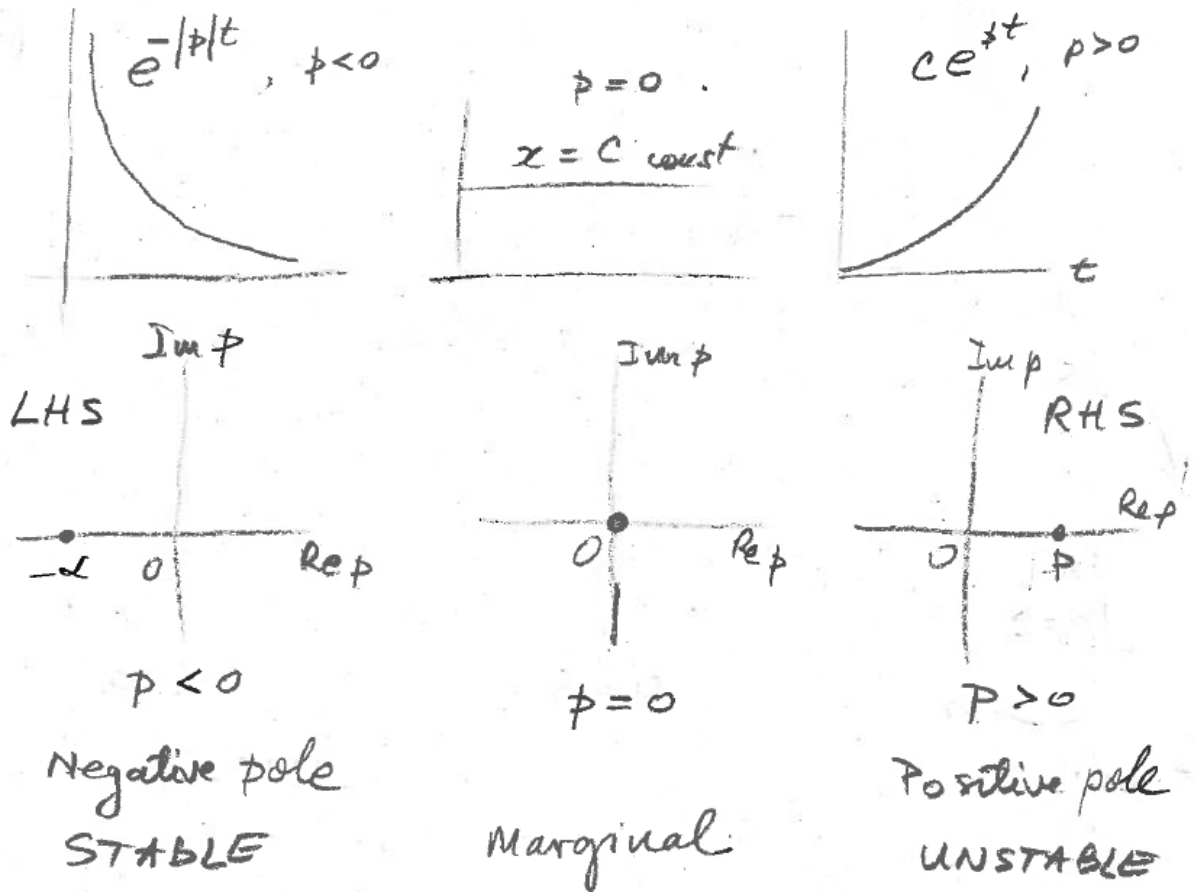
Figure 2.12: Free decay of a 1-DOF system

2.5.3 Stability of 1st-order systems

Using the general free response form eq 35 where p is a root of the characteristic equation (equation 33) and is called the “pole”. The stability of the system is dictated by the sign of p , i.e., its location in the complex p plane.

NOTE

A **Stable System** is a system where if a disturbance is applied, then the system returns to its initial state.

Figure 2.13: Stability of a first order system for different p values.

2.5.4 Stability of 1st-order Systems Under Forced Response

The total response of a first-order system is the sum of the complementary and particular solutions. From Equation 24, the total solution is

$$x(t) = x_c(t) + x_p(t). \quad (36)$$

For the first-order system considered here, the complementary solution is the free response,

$$x_c(t) = ce^{-t/\tau}. \quad (37)$$

Therefore, the total response can be written as

$$x(t) = ce^{-t/\tau} + x_p(t). \quad (38)$$

The constant c is determined by the initial condition, while the particular solution $x_p(t)$ is determined by the forcing function $f(t)$. Thus, the response of the system contains both a transient component associated with the initial condition and a forced component associated with the applied input.

In control theory, it is common to assume zero initial conditions when examining the input-output behavior of a system. Under this assumption, $x(0) = 0$ and the complementary response is eliminated, leaving

$$x(t) = x_p(t). \quad (39)$$

This distinction is important: stability is determined by the system's homogeneous response and therefore by its pole, while the particular response describes how the system responds to the applied forcing function.

2.6 2nd Order Systems

Consider the system:

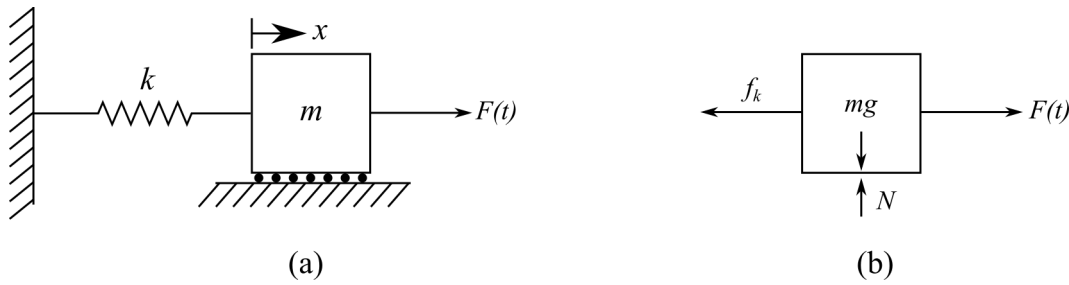


Figure 2.14: 1-DOF system with an external force ($F(t)$) applied, showing: (a) the system configuration; and (b) the free body diagram

Newton's second law of motion states:

$$m\ddot{x} = -kx + f^* \quad (40)$$

this can be rearranged to the standard equation of motion for a system with forced vibrations:

$$m\ddot{x}(t) + kx(t) = f^* \quad (41)$$

It is an 2nd-order inhomogeneous ordinary differential equation in time t . Similar to the solutions to the EOM for a 1st-order system, the solution for a 2nd-order system is made of two parts, the “complementary solution” and the “particular solution”. Again, this is written as:

$$x(t) = x_c(t) + x_p(t) \quad (42)$$

where $x_c(t)$ satisfies the homogeneous equation ($m\ddot{x}(t) + kx(t) = 0$) and represents the “free” response of the system. $x_p(t)$ satisfies the complete equation $m\ddot{x}(t) + kx(t) = f^*$ and represents the “forced” response of the system.

2.6.1 Homogeneous Equation (Free Vibration Response)

The homogeneous equation $x_c(t)$ is obtained by setting the forcing function of the EOM to zero:

$$m\ddot{x}(t) + kx(t) = 0 \quad (43)$$

normalizing by the mass returns:

$$\ddot{x}(t) + \frac{k}{m}x(t) = 0 \quad (44)$$

denoting the natural frequency of the system:

$$\omega_n^2 = \frac{k}{m} \quad (45)$$

or more simply (and traditionally)

$$\omega_n = \sqrt{\frac{k}{m}} \quad (46)$$

Inserting the natural frequency of the system back into the EOM yields the homogeneous equation in standard form:

$$\ddot{x} + \omega_n^2 x = 0 \quad (47)$$

To solve the homogeneous equation, assume the form:

$$x(t) = ce^{pt} \quad (48)$$

hence

$$\dot{x}(t) = pce^{pt} = px \quad (49)$$

$$\ddot{x}(t) = p^2 ce^{pt} = p^2 x$$

inserting these back into the EOM for the 2nd-order system results in:

$$p^2 x + \omega_n^2 x = 0 \quad (50)$$

this simplifies to the characteristic equation:

$$p^2 + \omega_n^2 = 0 \quad (51)$$

Rearranging this equations lead to

$$p^2 = -\omega_n^2 \quad (52)$$

and considering that $i = \sqrt{-1}$ to the solution

$$p_{1,2} = \pm i\omega_n \quad (53)$$

inserting these back into the EOM for the 2nd-order system results in:

$$x_c(t) = c_1 e^{i\omega_n t} + c_2 e^{-i\omega_n t} \quad (54)$$

which is the the complementary solution where c_1 and c_2 are constants to be determine. The complementary solution can also be written as:

$$x_c(t) = c \sin(\omega_n t + \phi) \quad (55)$$

where c is the amplitude of oscillation and ϕ is the phase angle.

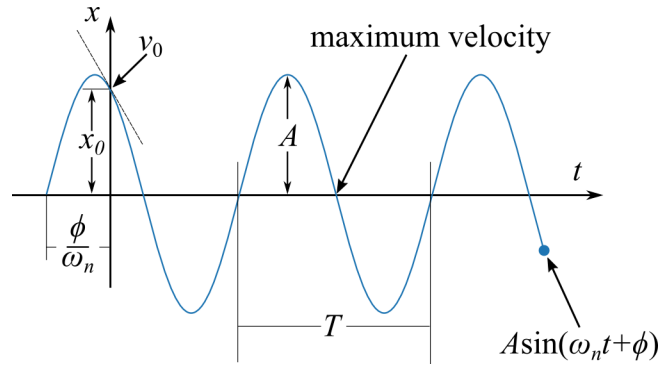


Figure 2.15: Summary of the temporal response for a 1-DOF system.

where x_0 and v_0 are the initial displacement and velocity at $t=0$ (i.e. the initial displacement).

Review 2.3 Euler's (pronounced oy-ler) formula, named after Swiss engineer and mathematician Leonhard Euler (1707-1783), is a mathematical formula in complex analysis that establishes the fundamental relationship between the trigonometric functions and the complex exponential function. Euler's formula states that for any real number x ,

$$e^{j\psi} = \cos(\psi) + j\sin(\psi) \quad (56)$$

where $j = \sqrt{-1}$. This equation can also be expressed as:

$$e^{-j\psi} = \cos(\psi) - j\sin(\psi) \quad (57)$$

This can be expressed in terms of polar coordinates as:

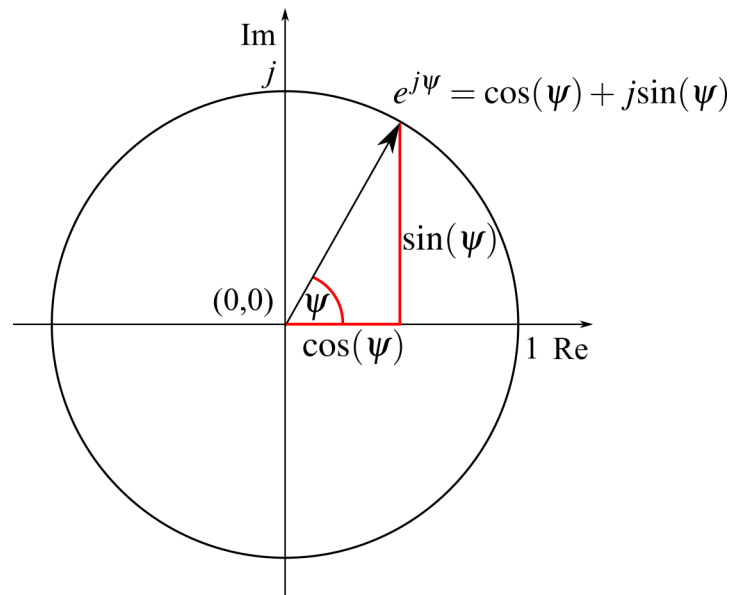


Figure 2.16: Euler's formula illustrated on the unit circle in the complex plane.

Proof

Showing that $x_c(t) = c_1 e^{i\omega_n t} + c_2 e^{-i\omega_n t}$ is the same as $x_c(t) = C \sin(\omega_n t + \phi)$, consider Euler's identity,

$$e^{i\alpha} = \cos(\alpha) + i \sin(\alpha) \quad (58)$$

therefore:

$$\begin{aligned} c_1 e^{i\omega_n t} &= c_1 \cos(\omega_n t) + c_1 i \sin(\omega_n t) \\ c_2 e^{-i\omega_n t} &= c_2 \cos(\omega_n t) + c_2 i \sin(\omega_n t) \end{aligned} \quad (59)$$

rearranging terms yields:

$$c_1 e^{i\omega_n t} + c_2 e^{-i\omega_n t} = (c_1 + c_2) \cos(\omega_n t) + (c_1 + c_2) i \sin(\omega_n t) \quad (60)$$

which is equivalent to:

$$c_1 e^{i\omega_n t} + c_2 e^{-i\omega_n t} = A \cos(\omega_n t) + B \sin(\omega_n t) \quad (61)$$

where:

$$A = c_1 + c_2; \quad B = i(c_1 - c_2) \quad (62)$$

Recall that $\sin(\alpha + \beta) = \sin(\alpha) \cos(\beta) + \cos(\alpha) \sin(\beta)$. Therefore, $x_c(t) = c \sin(\omega_n t + \phi)$ can be written as (with a little rearranging):

$$x_c(t) = c \cos(\phi) \sin(\omega_n t) + c \sin(\phi) \cos(\omega_n t) \quad (63)$$

Therefore we can show that:

$$A \cos(\omega_n t) + B \sin(\omega_n t) = c \sin(\omega_n t + \phi) \quad (64)$$

as

$$B = c \cos(\phi); \quad A = c \sin(\phi) \quad (65)$$

Moreover,

$$A^2 + B^2 = C^2 \sin^2(\phi) + C^2 \cos^2(\phi) = C^2 (\sin^2(\phi) + \cos^2(\phi)) = C^2 \quad (66)$$

and

$$C = \sqrt{A^2 + B^2}, \quad \frac{A}{B} = \frac{c \sin(\phi)}{c \cos(\phi)} = \tan(\phi), \quad \phi = \tan^{-1} \frac{A}{B} \quad (67)$$

The Standard form of Inhomogeneous Equation can be found, starting at the EOM:

$$m\ddot{x} + kx = f^* \quad (68)$$

divide by m , therefore:

$$\ddot{x} + \frac{k}{m}x = \frac{1}{m}f^* \quad (69)$$

Recall that $\frac{k}{m} = \omega_n^2$ and define the normalized forcing function as $f(t) = \frac{1}{k}f^*$, this results in:

$$\frac{1}{m}f^*(t) = \frac{1}{m}kf(t) = \omega_n^2 f(t) \quad (70)$$

inserting this back into the the EOM yields a 2nd-order inhomogeneous ODE in standard form:

$$\ddot{x} + \frac{k}{m}x = \omega_n^2 f(t) \quad (71)$$

2.7 Forced Vibration Response

The total vibration $x(t)$ of the system can be explained as:

$$x(t) = x_c(t) + x_p(t) = C \sin(\omega_n t + \phi) + x_p(t) \quad (72)$$

inserting the first part of this expression for $x(t)$ into equation 71 yields:

$$\ddot{x}_c + \ddot{x}_p + \omega_n^2(x_c + x_p) = \omega_n^2 f(t) \quad (73)$$

assuming that the complementary solution equals zero, this expression simplifies to:

$$\ddot{x}_p + \omega_n^2 x_p = \omega_n^2 f(t) \quad (74)$$

This expressions requires that we find c , ϕ , and $x_p(t)$; this is not easy.

2.8 Spring-mass-damper oscillator

Consider the system:

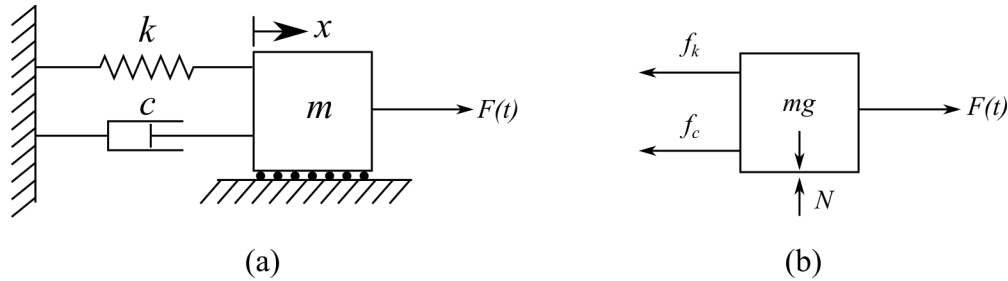


Figure 2.17: Damped 1-DOF system with an external force ($F(t)$) applied, showing: (a) the system configuration; and (b) the free body diagram

Newtons second law of motion tells us

$$m\ddot{x} = -kx - c\dot{x} + f^* \quad (75)$$

therefore, we can write the EOM as:

$$m\ddot{x}(t) + c\dot{x}(t) + kx(t) = f^*(t) \quad (76)$$

for the considered damped forced vibration problem. This is a 2nd order inhomogeneous ODE. As before, the solution of $x(t)$ for equation 76 is the sum of the complementary and particular solution where the complementary solution satisfies the homogeneous equation while the particular solution satisfies the inhomogeneous equation.

To solve the homogeneous equation, set the right hand side of equation 76 to zero to get the equation for damped free vibration:

$$m\ddot{x}(t) + c\dot{x}(t) + kx(t) = 0 \quad (77)$$

Again, assume $x(t) = Ce^{pt}$, therefore,

$$\begin{aligned} \dot{x}(t) &= pce^{pt} = px \\ \ddot{x}(t) &= p^2ce^{pt} = p^2x \end{aligned} \quad (78)$$

Putting these expressions back into the EOM yields:

$$mp^2x + cpx + kx = 0 \quad (79)$$

or the characteristic equation

$$mp^2 + cp + k = 0 \quad (80)$$

solving this expression leads to:

$$p_{1,2} = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m} = -\frac{c}{2m} \pm \sqrt{\frac{c}{2m} - \frac{k}{m}} \quad (81)$$

or:

$$p_{1,2} = -\frac{c}{2m} \pm i\sqrt{\frac{k}{m} - \left(\frac{c}{2m}\right)^2} \quad (82)$$

The critical damping value c_{cr} for the system is defined as the value of c that results in a 0 for the radicand (number under the square root). Therefore, we need:

$$\frac{k}{m} - \left(\frac{c_{cr}}{2m}\right)^2 = 0 \quad (83)$$

$$c_{cr} = \frac{km^2k}{m} = 4mk \quad (84)$$

this results in:

$$c_{cr} = 2\sqrt{mk} \quad (85)$$

The damping ratio ζ is the ratio between the damping c and the critical damping ratio c_{cr} , i.e.,

$$\zeta = \frac{c}{c_{cr}} = \frac{c}{2\sqrt{mk}} = \frac{c}{2m\omega_n} \quad (86)$$

Recall that the natural frequency of a system is $\omega_n = k/m$. Damping slightly reduces the natural frequency of the system, resulting in the damped natural frequency ω_d

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} \quad (87)$$

Using this expression for the damped natural frequency, the poles of the system can be identified as:

$$\begin{aligned} p_{1,2} &= -\zeta \omega_n \pm i\sqrt{\omega_n^2 - \zeta^2 \omega_n^2} \\ &= -\zeta \omega_n \pm i\sqrt{1 - \zeta^2} \\ &= -\zeta \omega_n \pm i\omega_d \end{aligned} \quad (88)$$

The damped free vibration response is:

$$x(t) = c_1 e^{(-\zeta \omega_n + i\omega_d)t} + c_2 e^{(-\zeta \omega_n - i\omega_d)t} \quad (89)$$

This simplifies to:

$$x(t) = e^{-\zeta \omega_n t} (c_1 e^{i\omega_d t} + c_2 e^{-i\omega_d t}) \quad (90)$$

or using Euler's equation,

$$x_c(t) = C e^{-\zeta \omega_n t} \sin(\omega_d t + \phi) \quad (91)$$

This equation breaks down into two parts, one for the decay of the system and one for the oscillations as shown below:

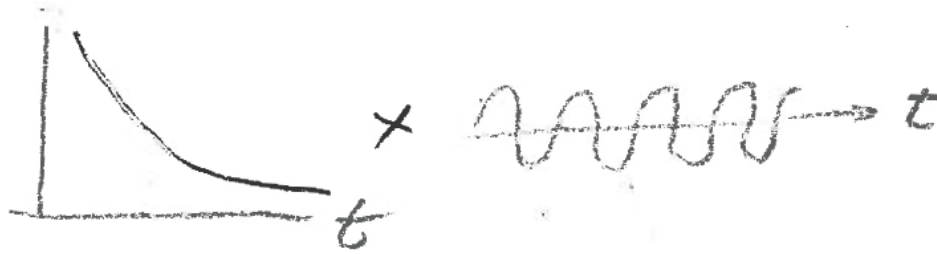


Figure 2.18: Decay and oscillations of the system.

Together, these make the free damped vibration response, as computed by the complementary solution $x_c(t)$

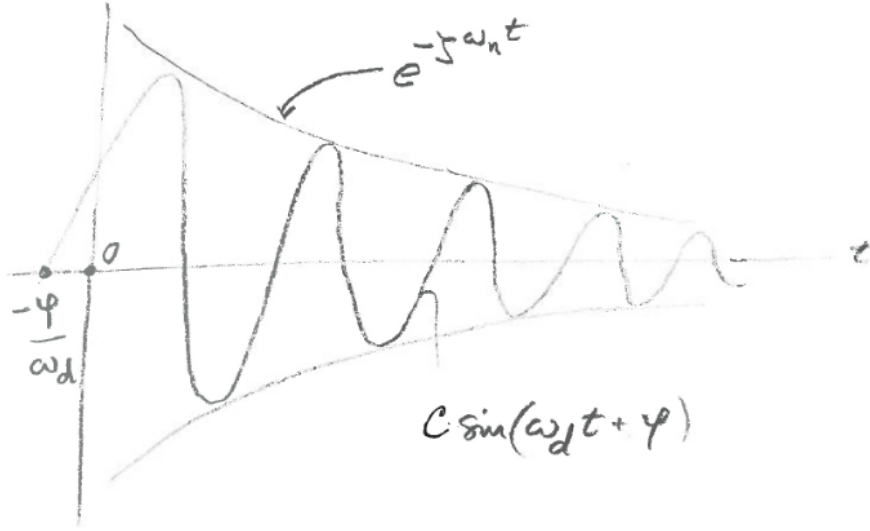


Figure 2.19: Free decay with oscillations.

Generating the standard form of the damped forced vibration problem is done by dividing the EOM by what leads the first term, mainly:

$$\ddot{x} + \frac{c}{m}\dot{x} + \frac{k}{m}x = \frac{1}{m}f^*(t) \quad (92)$$

The expression c/m can be rewritten as:

$$\frac{c}{m} = \frac{\zeta c_{cr}}{m} = \frac{\zeta 2\sqrt{mk}}{m} = 2\zeta \omega_n \quad (93)$$

while the previously defined normalized forcing function is $f(t) = \frac{1}{k}f^*(t)$, this results in a right hand side of EOM written as:

$$\frac{1}{m}f^*(t) = \frac{1}{m}kf(t) = \omega_n^2 f(t) \quad (94)$$

therefore, the standard form of the EOM for the 1-DOF damped forced vibration problem is written as a 2nd order ODE in standard form:

$$\ddot{x} + 2\zeta \omega_n \dot{x} + \omega_n^2 x = \omega_n^2 f(t) \quad (95)$$

The timed resolved solution for the forced vibration problem is:

$$\begin{aligned} x(t) &= x_c(t) + x_p(t) \\ &= e^{-\zeta \omega_n t} C \sin(\omega_d t + \phi) + x_p(t) \end{aligned} \quad (96)$$

Putting equation 96 back into equation 95 results in the expression:

$$(\ddot{x}_c + 2\zeta \omega_n \dot{x}_c + \omega_n^2 x_c) + \ddot{x}_p + 2\zeta \omega_n \dot{x}_p + \omega_n^2 x_p = \omega_n^2 f(t) \quad (97)$$

where we need to find C , ϕ , $x_p(t)$ based on the initial conditions of the system. This is not an easy task.

NOTE

The particular solution $x_p(t)$ depends on the form of the forcing function $f(t)$. For common forcing functions, $x_p(t)$ can be obtained using methods such as undetermined coefficients. The complementary solution describes the transient response, while the particular solution describes the forced response.

2.9 Stability of 2nd order systems

Recall that the damped free vibration response of a second-order system is given by

$$x_c(t) = Ce^{-\zeta\omega_n t} \sin(\omega_d t + \phi). \quad (98)$$

The damping ratio ζ determines how the system's response changes with time. For a positive damping ratio, the exponential term causes the amplitude of the response to decay, while the sinusoidal term produces the oscillation. The resulting stability characteristics can therefore be understood by examining both the time response and the locations of the system poles, as illustrated in Figure 2.20.

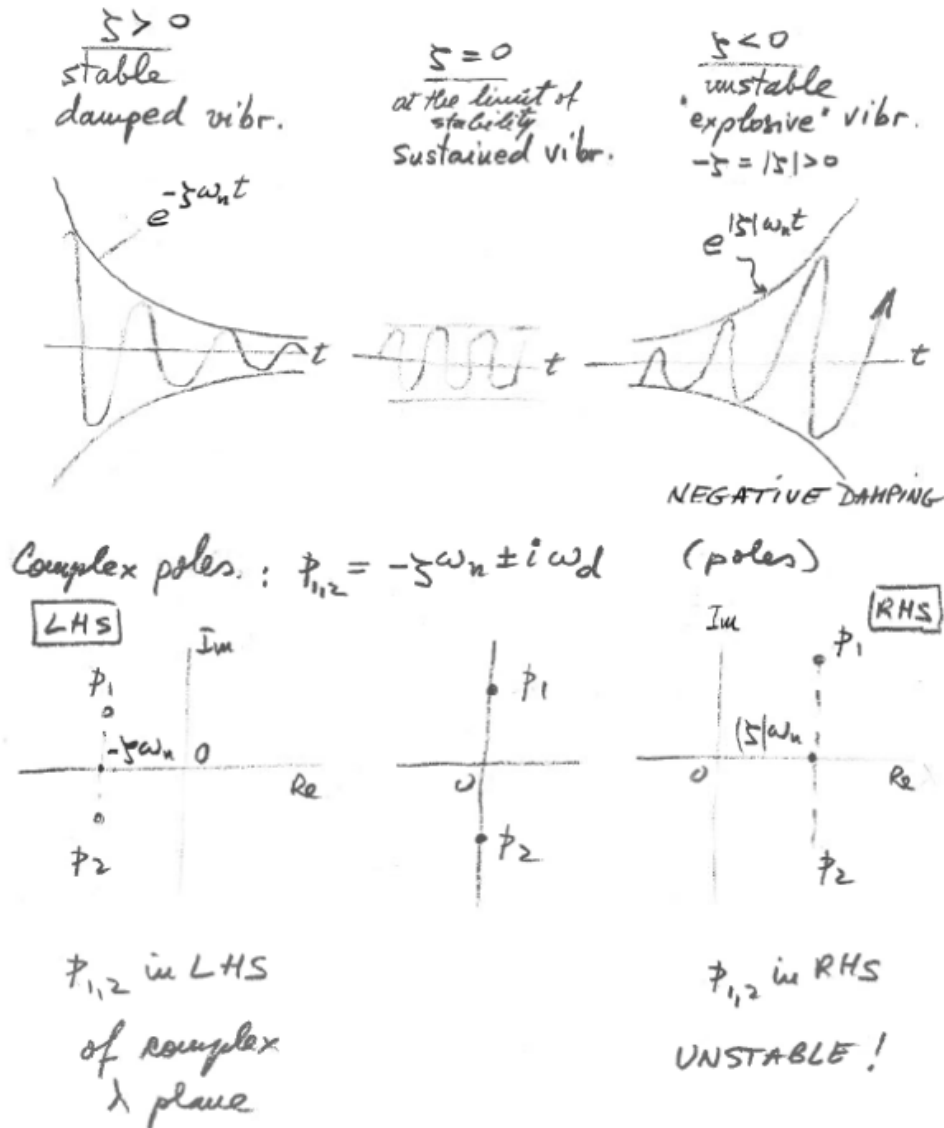


Figure 2.20: Stability of second-order systems for different damping conditions and pole locations.

For a second-order system, the poles are determined by the damping ratio and the natural frequency. From the characteristic equation, the pole locations are

$$p_{1,2} = -\zeta\omega_n \pm i\omega_n\sqrt{1-\zeta^2}. \quad (99)$$

The sign of the real part of the poles determines whether the vibration response decays or grows. When $\zeta > 0$, the real part is negative and the response decays with time. When $\zeta = 0$, the poles lie on the imaginary axis and the system exhibits sustained oscillations. When $\zeta < 0$, the real part is positive and the response grows with time, resulting in an unstable system.

The corresponding time responses for the three damping regimes are shown in Figure 2.21. For $0 < \zeta < 1$, the system is underdamped and exhibits oscillations with a decaying amplitude. The damped natural frequency is

$$\omega_d = \omega_n\sqrt{1-\zeta^2}. \quad (100)$$

At $\zeta = 1$, the system is critically damped and returns to equilibrium without oscillating. For $\zeta > 1$, the system is overdamped and also returns to equilibrium without oscillating, but more slowly than the critically damped case.

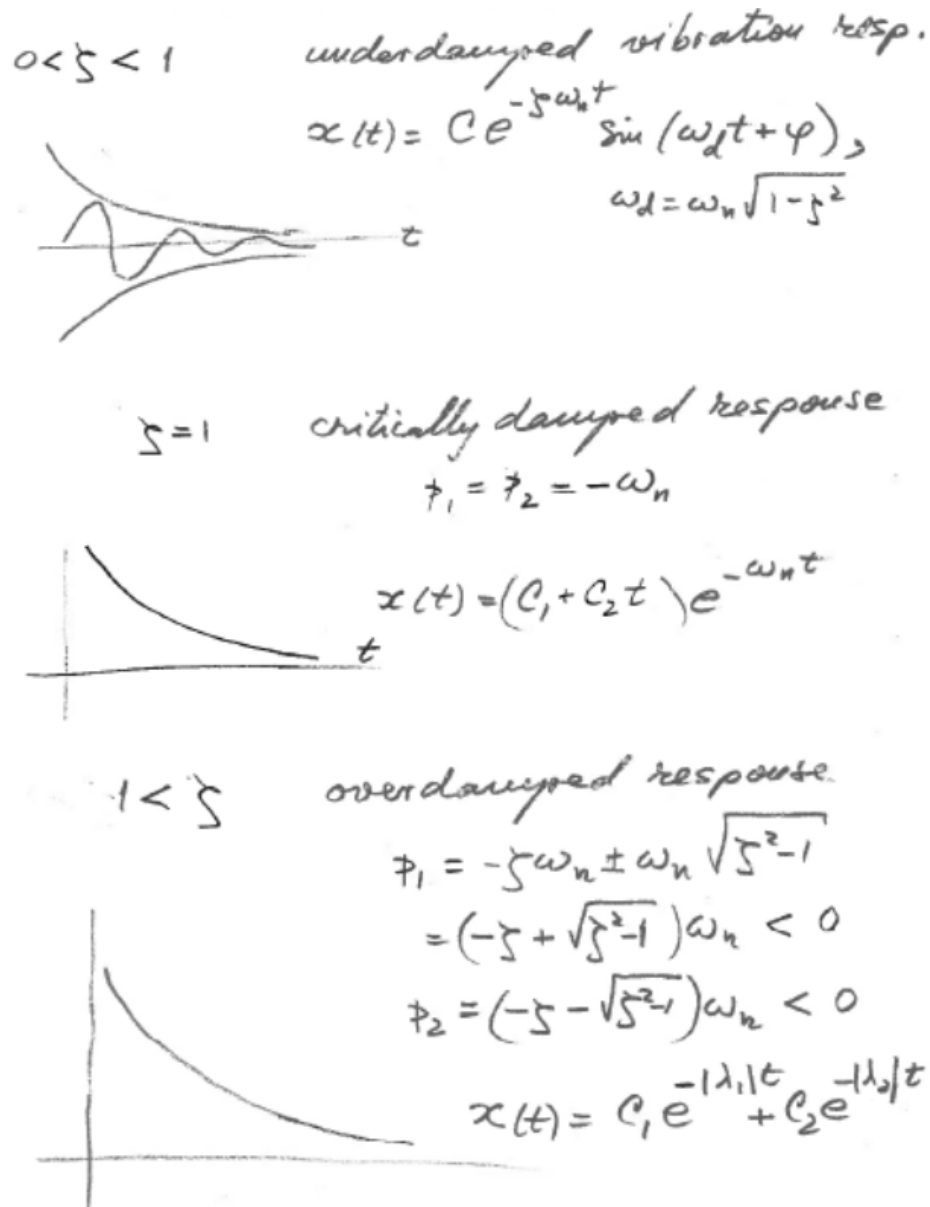


Figure 2.21: Time responses of underdamped, critically damped, and overdamped second-order systems.

The relationship between pole location and time response is summarized in Figure 2.22. Poles in the left half of the complex plane correspond to stable responses because their negative real parts produce exponential decay. Poles on the imaginary axis represent the limiting case of stability, producing sustained oscillations. Poles in the right half of the complex plane correspond to unstable responses because their positive real parts produce exponential growth.

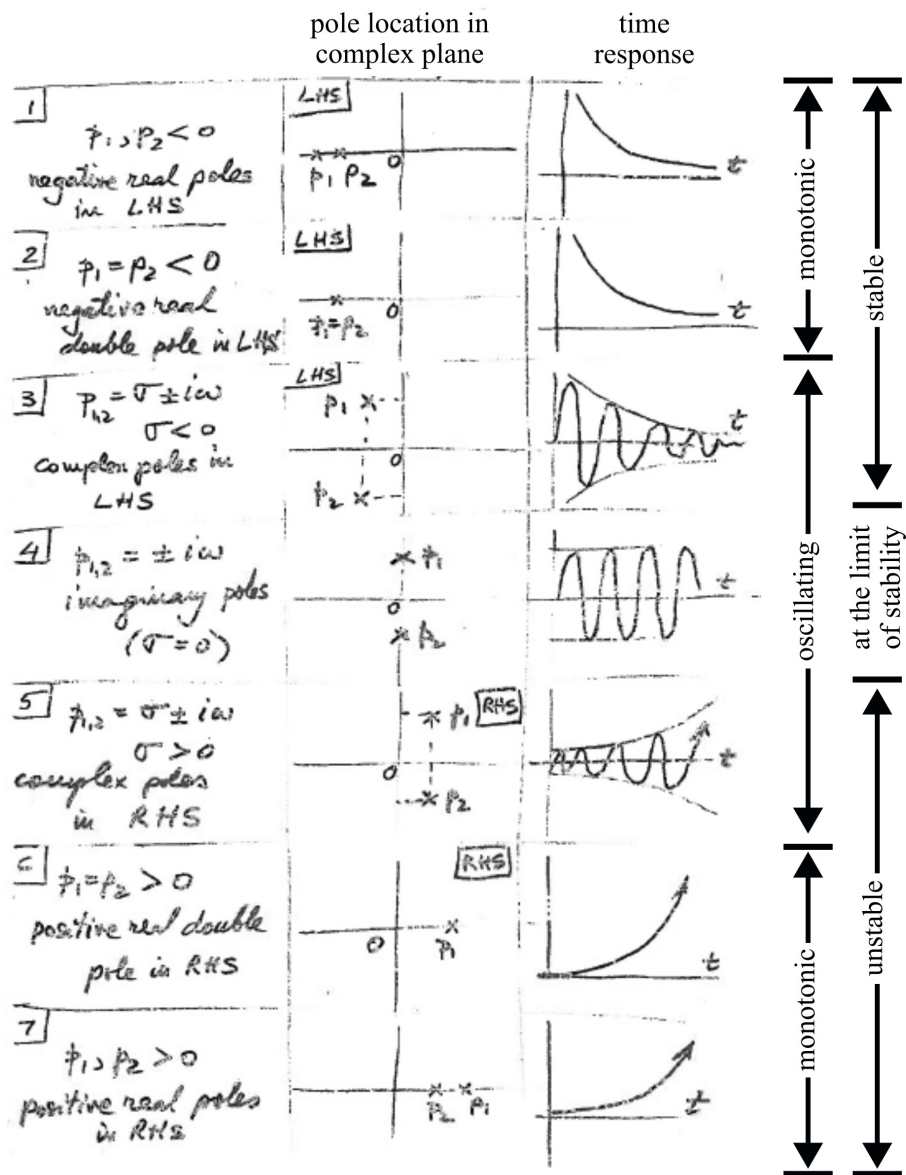


Figure 2.22: Relationship between second-order pole locations and corresponding time responses.

The pole locations therefore provide a direct way to evaluate the stability of a second-order system. A stable system has poles with negative real parts, while poles with positive real parts produce an unstable response.